

SANJAY KUMAR

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SUMMARY

Final-year Mechatronics Engineer with 7 months of commercial robotics experience at Meraque, building an RTK-GPS base station with ~2 cm rover gps accuracy and working on STM32/FreeRTOS firmware for autonomous AGVs. Built an NVIDIA Isaac Sim digital twin of a spraying AGV and designed a humanoid robot arm. Seeking Robotics Engineer or Embedded Systems Engineer roles.

EXPERIENCE

- Meraque Group Sdn Bhd** Malaysia
Electronic Engineering Intern February 2025 – September 2025
 - Built an RTK-GPS base station implementing FreeRTOS for multitask management, achieving ~2 cm positioning accuracy for autonomous AGV navigation.
 - Coordinated with the AI team to integrate perception outputs with low-level motor control over CAN/UART, closing the loop between detection and actuation for autonomous spraying.
 - Supported STM32 firmware development and hardware integration for spraying AGVs, contributing to field testing, customer demonstrations, and QA.
- Asia Pacific University Engineering Lab** Malaysia
Senior Technical Assistant (Part-time) November 2023 – Present
 - Maintain 30+ lab equipment (oscilloscopes, multimeters, power supplies, embedded kits) with near-zero downtime, resolving hardware faults to minimise lab disruption.
 - Supervise 20+ students per session, enforcing safety protocols and guiding correct equipment usage.

KEY PROJECTS

- Digital Twin of an Oil Palm Spraying AGV (Final Year Project)** *NVIDIA Isaac Sim, Gaussian Splatting*
 - Built a digital twin of an oil palm spraying AGV in NVIDIA Isaac Sim, enabling virtual pre-deployment testing without physical plantation visits.
 - Integrated a Gaussian Splatting-reconstructed plantation environment (3.32% mean geometric error), YOLOv8-seg tree detection, a precision sprayer controller, and real-time physical-virtual synchronisation
 - Validated through five structured tests: 90.5% sim-to-real sync accuracy, 96.3% tree detection rate, and 96.0% spray actuation accuracy in simulation
- Wall-E Companion Robot** *ROS2, CNN-BiGRU, DeepFace, Mistral LLM, Python*
 - Integrated a humanoid robot stack under a unified ROS2 architecture, fusing CNN-BiGRU children speech-emotion recognition (89% accuracy), DeepFace vision, and a Mistral LLM for real-time child interaction.
 - Engineered a neck mechanism (DC motor, worm gear, rotary encoder) with a P-controller for closed-loop face tracking.

LEADERSHIP

- IMechE APU Student Chapter** Malaysia
Vice-President August 2024 – May 2026
 - Led a committee of 10+ members and directed planning and execution of 30+ technical workshops, competitions, and networking events.

EDUCATION

- Asia Pacific University** Kuala Lumpur, Malaysia
BEng (Hons) Mechatronic Engineering 2021 – 2026

TECHNICAL SKILLS

- Languages:** Python, C++, C#, PLC Programming.
- Robotics & Embedded:** ROS2, STM32, FreeRTOS, RTK-GPS, CAN/UART, Isaac Sim, YOLOv8, Gaussian Splatting.
- Engineering Software & Tools:** MATLAB/Simulink, SolidWorks, LabVIEW, KiCad, LTSpice, Altair, Git, Linux.
- Systems:** Hardware-Software Integration, Sensor Fusion, System Validation, QA, Technical Documentation.
- Spoken:** English (Fluent), Urdu (Native), Hindi (Fluent).